Computer Networks Module 1 Assignment

Topic: How the Internet Delivers a Web Page (Example: YouTube)

1. TCP/IP Protocol Stack

Application Layer HTTP/HTTPS, DNS, FTP
Transport Layer TCP/UDP
Internet Layer IP
Network Access Layer Ethernet/Wi-Fi

Application layer provides network services to applications. Transport layer uses TCP/UDP. Internet layer routes packets using IP. Network Access layer transmits frames over Ethernet or Wi■Fi.

2. Role of DNS

DNS converts a domain name such as www.youtube.com into an IP address. The browser queries a DNS resolver, receives the server IP address, and then connects to the destination server.

3. HTTP

After the TCP connection is established, the browser sends an HTTP GET request. The server processes it and returns an HTTP response containing HTML, CSS, JavaScript and other resources.

4. TCP Reliability

TCP provides reliable communication using the three-way handshake, sequence numbers, acknowledgements, retransmissions, flow control and error checking.

5. Architecture

YouTube follows the Client–Server architecture because browsers (clients) request services from YouTube servers that store and deliver webpages and videos.

6. HTTP vs FTP

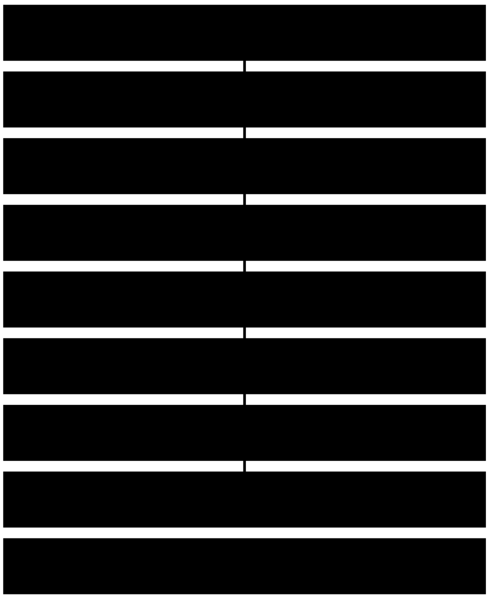
HTTP	FTP
Used for web pages	Used for file transfer
Port 80/443	Ports 20/21

Stateless	Session-oriented
Transfers web resources	Transfers files

7. Cookies

Cookies are small files stored by the browser. They remember login sessions and preferences. Example: YouTube keeps users signed in and recommends videos based on previous activity.

8. Communication Flow Diagram



9. Conclusion

Application-layer protocols enable everyday Internet communication. DNS resolves names into IP addresses, HTTP exchanges webpages, TCP ensures reliable delivery, and cookies improve usability by remembering user information. Together these technologies make web browsing fast, secure and reliable, allowing users to access online services efficiently.